F72-SR-001
415V REAGENTS SWITCHROOM

F72-SW-072
415V REAGENTS 415V MOTOR CONTROL CENTRE

CONTINUED ON SHT 1
2000-F72-SLD-EL-10201

CONTINUED ON SHT 3
2000-F72-SLD-EL-10203

NORTHERN STAR

Approved	X
Approved as Noted	
Not Approved	
For Information Only	

Signed [redacted] Date 10/12/2025

ACCEPTANCE DOES NOT RELIEVE THE SUPPLIER / CONTRACTOR OF ITS RESPONSIBILITY OR LIABILITIES UNDER THE PURCHASE ORDER / CONTRACT

415V 3NE~ 50Hz 2500A 65kA for 1sec

1F-13	3R-01	3R-05	3R-09	3R-13	3R-17	3R-21	4R-01	4R-05	4R-09	4R-13	4R-17
400A	160A	160A	160A	160A	160A	160A	160A	160A	160A	160A	160A
2A 32A	2A 32A	2A 32A	2A 32A	2A 32A	2A 32A	2A 32A	2A 32A	2A 32A	2A 32A	2A 32A	2A 32A
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250A	100/1A	100/1A	100/1A	100/1A	100/1A	100/1A	100/1A	100/1A	100/1A	100/1A	100/1A
EOL											
1											
110kW	55kW	55kW	55kW	55kW	55kW	55kW	55kW	55kW	55kW	55kW	55kW
F72-HT-491 CAUSTIC STORAGE TANK IMMERSION HEATER 2000-F72-SCM-EL-10405	F72-MB-411A LIME SLAKING MILL 1 MOTOR 1 2000-F72-SCM-EL-10670	F72-MB-411B LIME SLAKING MILL 1 MOTOR 2 2000-F72-SCM-EL-10671	F72-MB-411C LIME SLAKING MILL 1 MOTOR 3 2000-F72-SCM-EL-10672	F72-MB-411D LIME SLAKING MILL 1 MOTOR 4 2000-F72-SCM-EL-10673	F72-MB-411E LIME SLAKING MILL 1 MOTOR 5 2000-F72-SCM-EL-10674	F72-MB-411F LIME SLAKING MILL 1 MOTOR 6 2000-F72-SCM-EL-10675	F72-MB-412A LIME SLAKING MILL 2 MOTOR 1 2000-F72-SCM-EL-10676	F72-MB-412B LIME SLAKING MILL 2 MOTOR 2 2000-F72-SCM-EL-10677	F72-MB-412C LIME SLAKING MILL 2 MOTOR 3 2000-F72-SCM-EL-10678	F72-MB-412D LIME SLAKING MILL 2 MOTOR 4 2000-F72-SCM-EL-10679	F72-MB-412E LIME SLAKING MILL 2 MOTOR 5 2000-F72-SCM-EL-10680

NOTES

1. EARTH-NEUTRAL LINK SHALL BE INSTALLED IN INCOMER TIER. SIGNAGE SHALL BE PLACED ON MCC AND ON TRANSFORMER READING "M.E.N. LINK LOCATED IN F72-SW-072 INCOMER TIER" CABLE SHALL BE IDENTIFIED C/W LABEL STATING "M.E.N. LINK - DO NOT REMOVE".
2. SWITCHBOARD SHALL BE EXTENSIBLE AND BE PROVIDED WITH THE FACILITY TO EXTEND THE MAIN BUSBAR AND APPEND ADDITIONAL TIERS AT ONE END WITHOUT MAJOR STRUCTURAL DISASSEMBLY.
3. UPSTREAM TRANSFORMER FEEDER SHALL BE TRIPPED ON ACTIVATION OF SWITCHROOM EMERGENCY PUSHBUTTON, DETECTION OF EARTH FAULT OR ARC FAULT DETECTION.
4. SWITCHBOARD VENDOR SHALL SELECT SWITCHGEAR TO PROVIDE TYPE-2 CO-ORDINATION SUITABLE FOR HIGH EFFICIENCY MOTORS.
5. ALL DRIVE I/O VIA COMMS NETWORK TO DCS. FOR COMMUNICATIONS BLOCK DIAGRAM REFER 2000-F72-BLK-IC-10001.
6. SURGE DIVERTERS SHALL PROVIDE FEEDBACK TO DCS VIA VOLT-FREE CONTACTS. SURGE DIVERTER FUSES SHALL BE SIZED BY VENDOR TO SUIT DUTY.
7. MINIATURE CIRCUIT BREAKERS SHALL BE SIZED AND RATED TO SUIT ACB AND INCOMER REQUIREMENTS.
8. THERMAL OVERLOAD UNITS MAY BE SUBSTITUTED WITH ELECTRONIC (MAGNETIC) OVERLOAD UNITS TO SUIT DUTY.
9. TOLs TO BE MOUNTED IN PURPOSE-BUILT ENCLOSURE BENEATH VSD. TOLs MAY BE INCLUDED IN MCC MODULE WITH ENGINEERING APPROVAL.
10. ANY DRIVE RTDs SHALL BE WIRED TO DEDICATED REMOTE I/O CARDS IN THE FIELD. RTD RELAYS MAY BE INCLUDED IN THE MCC MODULE WITH ENGINEERING APPROVAL.
11. CIRCUIT BREAKER TO BE SUBSTITUED FOR FUSED-SWITCH WHERE REQUIRED BY EQUIPMENT MANUFACTURERS GUIDELINES.
12. EMERGENCY SHUTDOWN (E-SHUT) PUSH BUTTON WIRED TO SHUNT TRIP NOMINATED CIRCUIT BREAKERS.

LEGEND

- DIRECT ON-LINE MOTOR STARTER GENERIC - NOTE 4, 5
- DIRECT ON-LINE MOTOR STARTER REVERSING - NOTE 4, 5
- EOL — ELECTRONIC OVERLOAD UNIT NOTE 4, 5
- THERMAL OVERLOAD UNIT NOTE 8
- VARIABLE SPEED DRIVE NOTE 4, 5
- /> — EARTH LEAKAGE RELAY NOTE 4, 5
- FIELD-MOUNTED DECONTACTOR
- JB — JUNCTION BOX
- t° — MOTOR THERMISTOR (ONLY REQ. FOR MOTORS >37kW - 250kW)
- RTD — MOTOR RTD (ONLY REQ. FOR MOTORS >250kW)
- MOTOR HEATERS (ONLY REQ. FOR MOTORS >90kW)
- 49 63 71 — TRANSFORMER PROTECTIVE DEVICES (REFER ANSI STANDARDS FOR DESCRIPTIONS)
- CIRCUIT BREAKER (NOTE 11)
- xxxA / xxxA — ST TRIP SET POINT / CB SIZE
- AFD — ARC FLASH DETECTION
- ECR — EARTH CONTINUITY RELAY
- E-SHUT — SWITCHROOM EMERGENCY SHUTDOWN
- 63A 3 PHASE SOCKET OUTLET

2000-F72-SLD-EL-10202_Rev1

F72-SW-072 415V MCC - single line diagram sheet 2

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